

Differentiating Power Series: If

All of the above only work within radius of convergence

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$$

Just a power series centered at some a

has radius of convergence $R > 0$, then it is differentiable on the interval $(a-R, a+R)$ with

Converges for $-R < x < R$
 ↳ Endpoints will depend, no?

$$f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}$$

Interval based on ROC & center

↳ When you diff you lose a term → index forward

↳ Derivative of a power series is another power series

Integrating Power Series: If

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$$

has radius of convergence $R > 0$, then the indefinite integral of $f(x)$ is

$$\int f(x) dx = \sum_{n=0}^{\infty} \frac{c_n}{n+1} (x-a)^{n+1} + C$$

↳ When you integrate you add a term with a positive index. Need C to account for proper index. Always add term $+C$.

If $[s, t]$ is fully contained in the interval of convergence, then the definite integral is

$$\int_s^t f(x) dx = \sum_{n=0}^{\infty} c_n \int_s^t (x-a)^n dx = \left(\sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1} \right) \Big|_a^b$$

Only works when bounds

are within radius of conv.

$$(-R < s \text{ and } t < R)$$

Example: Consider $f(x) = \sum_{n=0}^{\infty} (-1)^n x^n$

(a) Find the radius and interval of convergence.

$$L = \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} x^{n+1}}{(-1)^n x^n} \right| = \left| \frac{-x}{1} \right| = |x|$$

$$\sum_{n=0}^{\infty} (-1)^n x^n \text{ converges } -1 < x < 1$$

When $x=1 \rightarrow \sum_{n=0}^{\infty} (-1)^n \rightarrow$ Diverges by divergence test

When $x=-1 \rightarrow \sum_{n=0}^{\infty} (-1)^n \cdot (-1) \rightarrow$ Diverges by divergence test
 $= \sum_{n=0}^{\infty} (-1)^{n+1}$

$$\text{I.O.C.} = (-1, 1)$$

(b) Find $f'(x)$

For $|x| < 1$ & R.O.C.

You don't have to expand out, you can just use the definition b/c you can't fully expand an inf. series

$$\begin{aligned} f(x) &= 1 - x + x^2 - x^3 + \dots \\ f'(x) &= 0 - 1 + 2x - 3x^2 + \dots = \sum_{n=1}^{\infty} (-1)^n n x^{n-1} \end{aligned}$$

index + 1

(c) Find $\int f(x) dx$

When $|x| < 1$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1} + C = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} + C$$

Don't forget!

Index stays the same because you add the +1

(d) Find $\int_0^2 f(x) dx$

Can't be done b/c $[0, 2]$ is not contained in $(-1, 1)$

What about $\int_0^{1/2} f(x) dx$?

Can be done in R.O.C.

$$\int_0^{1/2} f(x) dx = \left(\sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{x}{2}\right)^{n+1}}{n+1} \right) \Big|_0^{1/2} = \left(\sum_{n=0}^{\infty} (-1)^n \frac{0^{n+1}}{n+1} \right)$$

One approach to find power series representation of $f(x)$:

$$\text{Use } \frac{a}{1-r} = \underbrace{\sum_{n=0}^{\infty} ar^n}_{\text{Geom. series}} \text{ for } |r| < 1$$

↳ we already know interval for this is $(-1, 1)$

Example: Find a power series representation for $f(x) = \frac{1}{2+x}$

Example: Find a power series representation for $f(x) = \frac{1}{1-x^3}$

Hint: In the previous class, we found $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, for $|x| < 1$.

Let $r = x^3$, $q = 2 \rightarrow$ Use $\frac{q}{2-r} = \sum_{n=0}^{\infty} q r^n$ for $|r| < 2$
 $\downarrow$
 1 ∞ ∞
 $\hookrightarrow$ we already know interval for this is $(-2, 2)$

$$\frac{1}{1-x^3} = \sum_{n=0}^{\infty} (x^3)^n = \sum_{n=0}^{\infty} x^{3n} \text{ for } |x^3| < 1$$
$$\geq |x| < 1^{\frac{1}{3}}$$
$$= |x| < 1$$

$$f(x) = \sum_{n=0}^{\infty} x^{3n} \text{ for } |x| < 1$$

Example: Find a power series representation for

$$f(x) = \frac{x}{1+9x^2}$$

Use $\frac{q}{1-r} = \sum_{n=0}^{\infty} ar^n$ for $|r| < 1$
 Geom. series

we already know interval for this is $(-1, 1)$

we want to get this to look like $\frac{q}{1-r}$

$$= x \cdot \frac{1}{1+9x^2}$$

we will multiply this back in

$$= x \cdot \frac{1}{1-(-9x^2)}$$

$\frac{1}{1-(-9x^2)} = \sum_{n=0}^{\infty} (-9x^2)^n$ conv. for $| -9x^2 | < 1$
 $| -9x^2 | < 1$
 $|x|^2 < \frac{1}{9}$
 $|x| < \sqrt{\frac{1}{9}}$
 $|x| < \frac{1}{3}$
 $R = \frac{1}{3}$

$$f(x) = x \cdot \frac{1}{1-(-9x^2)} = x \left(\sum_{n=0}^{\infty} (-9x^2)^n \right) = \sum_{n=0}^{\infty} (-9)^n (x^{2n+1})$$

If it converges, you can multiply this back in like it's a constant w/o x's just a var

$$= \sum_{n=0}^{\infty} (-9)^n x^{2n+1}$$

for $|x| < \frac{1}{3}$

$$f(x) = \sum_{n=0}^{\infty} (-9)^n x^{2n+1} \quad \text{for } |x| < \frac{1}{3}$$

Example: Find a power series representation for $f(x) = \frac{5x}{(1-x)^2}$

$$f(x) = 5x \cdot \frac{1}{(1-x)^2}$$

We want smth that looks like $\frac{1}{1-x}$

Note that $\frac{d}{dx} \left(\frac{1}{1-x} \right) = \frac{d}{dx} (1-x)^{-1} = -1(1-x)^{-2} \cdot -1 = \frac{1}{(1-x)^2}$

$$= 5x \cdot \frac{d}{dx} \left(\frac{1}{1-x} \right)$$

$$= 5x \cdot \frac{d}{dx} \left(\sum_{n=0}^{\infty} x^n \right)$$

$a = 1$
 $r = x$
 Converges for $|x| < 1$

$$= 5x \left(\sum_{n=1}^{\infty} nx^{n-1} \right)$$

$$= \sum_{n=1}^{\infty} 5nx^{n-1}$$

$$= \sum_{n=1}^{\infty} 5n x^n \text{ for } |x| < 1$$

$$f(x) = \sum_{n=1}^{\infty} 5n x^n \text{ for } |x| < 1$$

Use $\frac{a}{1-r} = \sum_{n=0}^{\infty} ar^n$ for $|r| < 1$
 Geom. series

↳ we already know interval for this is $(-1, 1)$

Example: Find a power series representation for $f(x) = \tan^{-1} x$

1. Find power series for $\frac{1}{1+x^2}$

2. Integrate it

3. Use $\arctan(0) = 0$

1. $\frac{1}{1+x^2}$

$q=1$ $r=-x^2$

Use $\frac{q}{1-r} = \sum_{n=0}^{\infty} q r^n$ for $|r| < 1$

$\frac{1}{1-x^2} = \sum_{n=0}^{\infty} (-x^2)^n$ for $|x| < 1$

$\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$ for $|x| < 1$

$\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$ for $|x| < 1$

$\arctan x = \int \frac{1}{1+x^2} dx$

2. $\arctan(x) = \int \frac{1}{1+x^2} dx = \int \sum_{n=0}^{\infty} (-1)^n x^{2n} dx = \sum_{n=0}^{\infty} (-1)^n \int x^{2n} dx$

$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+2}}{2n+2} + C$

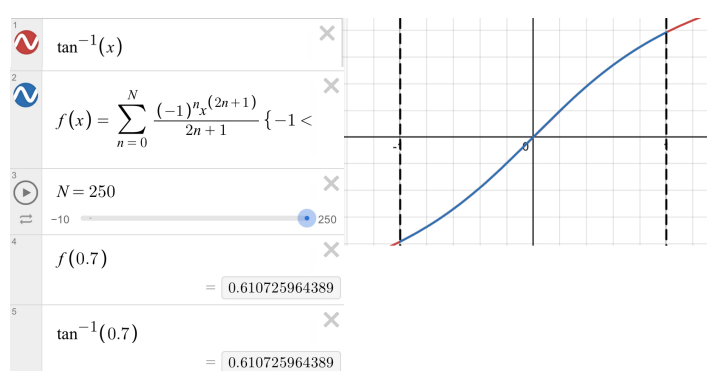
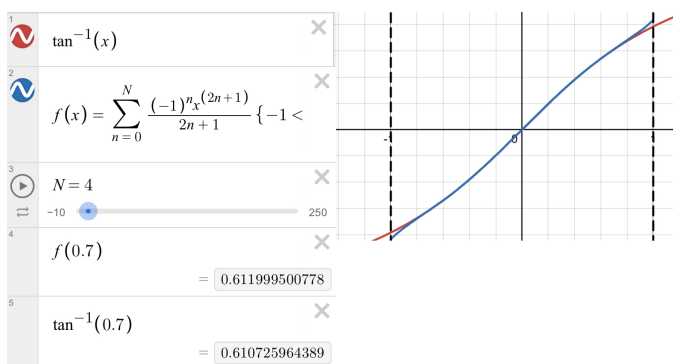
3. $\arctan(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+2}}{2n+2} + C =$

$C + x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$

$0 = \arctan(0) = C \rightarrow C = 0$

So $\arctan(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+2}}{2n+2}$

for $|x| < 1$



With a larger and larger partial sum you get closer and closer to the actual function, and this series is much easier to work with than inverse tan since it's just a polynomial